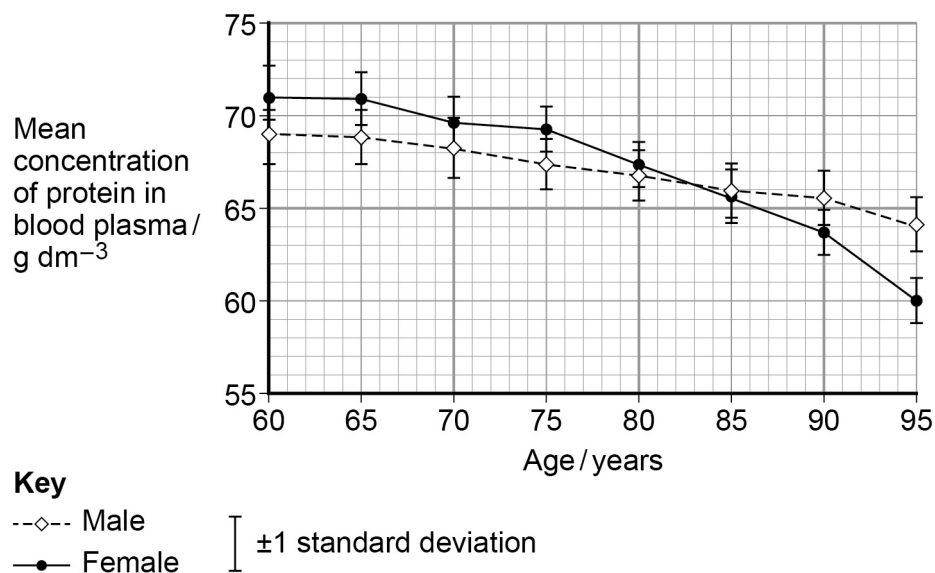


0 4

Scientists investigated how the concentration of protein in blood plasma changes in people between the ages of 60 and 95.

Figure 4 shows the scientists' results. The bars show ± 1 standard deviation.

Figure 4



0 4 . 1

What is the difference between males and females in the fall in mean concentration of protein in blood plasma between 60 and 95 years?

[1 mark]

Answer = _____ g dm^{-3}



0	4	.	2
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Use **Figure 4** to calculate the rate of change of the mean concentration of protein in the blood plasma of males between the ages of 60 and 95.

Show your working.

[2 marks]

Answer = _____ g dm⁻³ year⁻¹

0	4	.	3
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What can you conclude from **Figure 4** about the effect of ageing on the mean concentration of protein in the blood plasma in males and females?

[2 marks]

Question 4 continues on the next page

Turn over ►



0	4	.	4
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The scientists measured the absorption of each sample of blood plasma using a colorimeter. They used a calibration curve to find the concentration of protein in samples of blood plasma.

Describe how the scientists could obtain data to produce a calibration curve and how they would use the calibration curve to find the concentration of protein in a sample of blood plasma.

[3 marks]

[Extra space]

0	4	.	5
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Older people are more likely to suffer from infectious diseases.

Suggest how this may be linked to the decrease in the mean concentration of protein in the blood as people get older.

[1 mark]

